SCIENTIFIC DATA VISUALIZATION

Who really rents in Milan on airbnb?

Sofia Cipolloni · Laura Ferri

11 November 2026



Inside Airbnb · Milan · 21 September 2026

THE DATA

Dataset

Source: Inside Airbnb; every Milan Airbnb listing on 21 September 2026

OBSERVATIONS

25,338 listings in the file
11,475 hosts

↓ − 81 shared or hotel rooms

25,257 homes and private rooms

↓ − 3,666 not bookable on 21 September (no price)

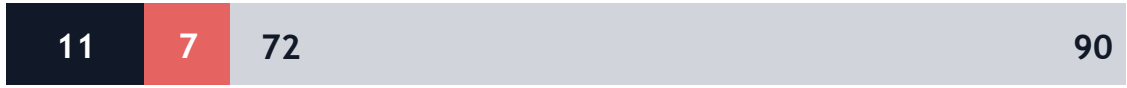
21,591 with a price on the day

↓ − 81 prices below 20 € or above 2,000 € (not plausible)

21,510 listings analyzed

84.9% of the file is kept

VARIABLES



11 used to find the types *the only ones the clustering sees*

- **Space** guests, bedrooms, beds, bathrooms, amenities
- **Price** € per night
- **Use** minimum stay, days open, reviews in 12 months
- **Host** how many listings, years on Airbnb

7 kept aside to test them *never shown to the clustering*

- **Trust** Superhost badge, guest rating
- **Licence** CIN code
- **Location** latitude, longitude, neighbourhood
- **Room** entire home or private room

THE PLAN

Six questions

Q1	Do Milan listings fall into distinct types?	PCA · hierarchical clustering · k-means
Q2	What is each type like?	profile cards
Q3	Do the types differ in hosting quality and licence compliance?	hypothesis tests
Q4	Are the types spread differently across Milan?	maps · distance · price
Q5	Can the listing alone predict a Superhost?	logistic regression vs neural network
Q6	Is the market growing and who drives it?	time series · calendar heatmap

THE METHOD

Ten steps in four phases

PREPARE

1 Clean
filter rows, read text fields, log scale, z-scores

2 Explore
correlations and distributions

FIND THE TYPES

3 PCA
11 measures → 4 components

4 Hierarchical clustering
4 linkages; Ward suggests k

5 K-means
k = 4, 100 starts; silhouette vs shuffled data

6 Bootstrap
stability of both methods, agreement

TEST THE TYPES

7 Profiles
what each type is like

8 Hypothesis tests
on the hidden variables: χ^2, Kruskal–Wallis, effect sizes, adjusted odds, maps

PREDICT & LOOK BACK

9 Predict Superhost
logistic vs neural network, split by host, SHAP

10 Over time
new listings by month and by host size

One rule everywhere: listings of the same host are not independent, so tests cluster errors by host and the train/test split keeps each host on one side.

Exploratory Data Analysis

Q1

Correlations between the 11 measures

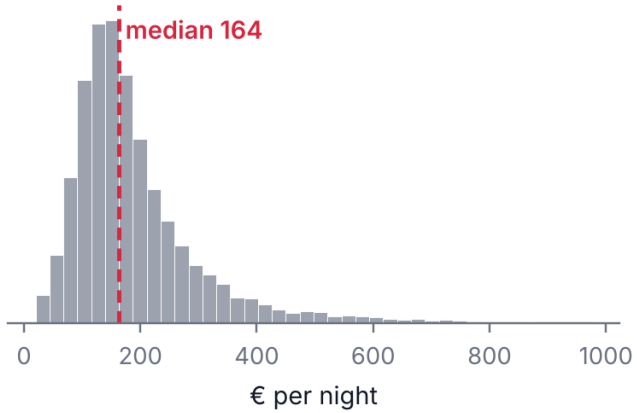


+0.78 guests, beds and bedrooms move together

-0.41 more reviews where the minimum stay is short

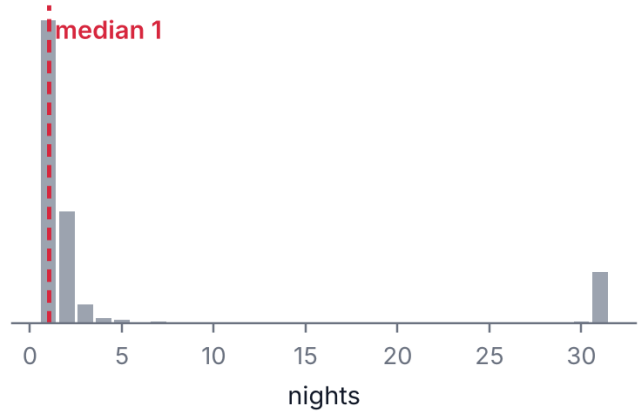
Distributions of two measures with long right tails

Price per night



Median 164 €; two in three cost 100–250 €,
4% over 500 €.

Minimum nights



86% ask 1–3 nights;
12% ask 28 or more: the monthly lets.

Long right tails → log scale before PCA, so a few huge values do not dominate the distances.



Do Milan listings fall into distinct types?



Q1

PCA - Principal Component Analysis

66%

of the information kept by 4 components
(eigenvalues > 1)

PC1	29%	**Size and price**
PC2	16%	**Busy short stays**
PC3	12%	**Professional host**
PC4	9%	**Always available**

	PC1	PC2	PC3	PC4
Guests	+0.49	-0.04	-0.02	-0.01
Bedrooms	+0.46	-0.18	-0.14	-0.12
Beds	+0.46	-0.11	-0.08	-0.13
Bathrooms	**+0.38**	-0.20	-0.07	-0.12
Price	+0.33	+0.02	+0.21	+0.45
Min nights	-0.15	**-0.53**	-0.34	-0.22
Days available	+0.00	-0.24	-0.09	**+0.71**
Reviews 12 m	+0.11	**+0.55**	+0.04	-0.18
Host listings	+0.07	-0.11	**+0.72**	+0.09
Years as host	-0.01	-0.27	**+0.51**	**-0.39**
Amenities	+0.20	**+0.44**	-0.14	-0.03

Q1

Three steps to find the types

1

Hierarchical clustering

How many groups?

→

2

K-means

Are the groups real?

→

3

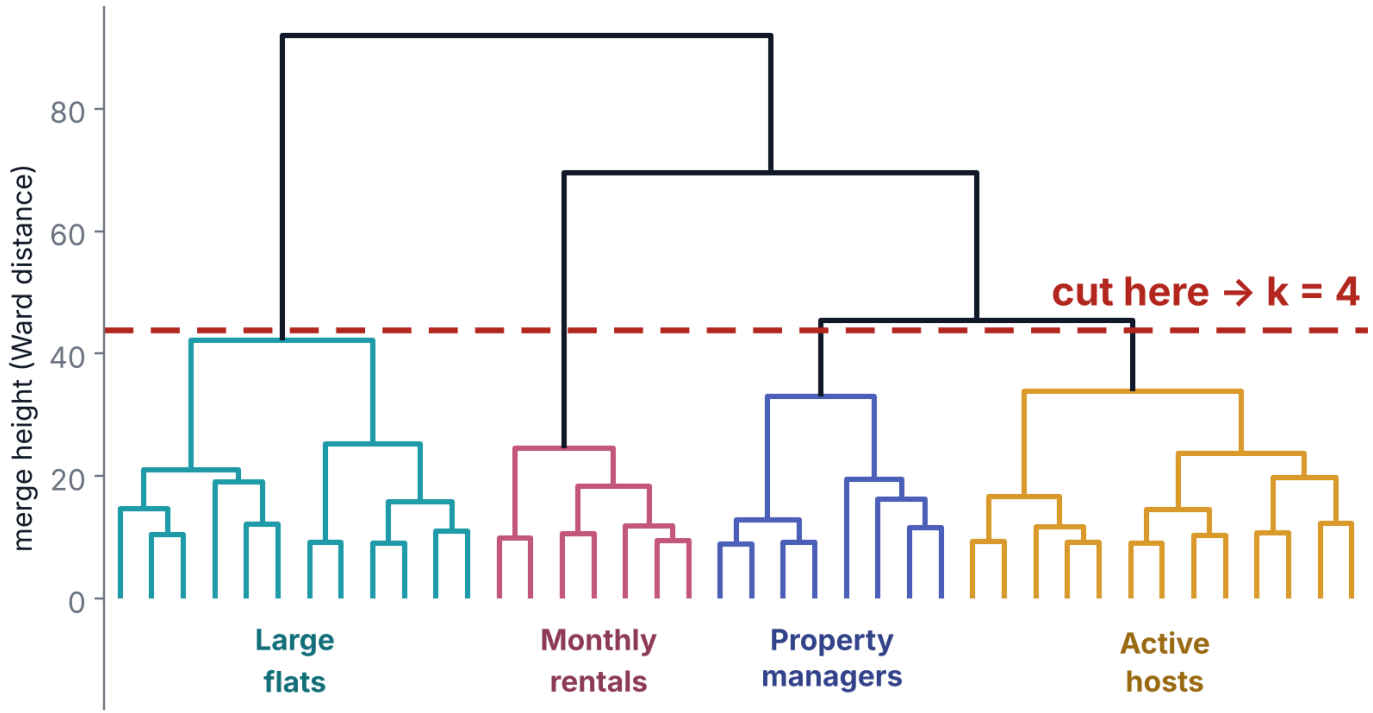
Comparison · bootstrap

Are the groups stable?

Q1

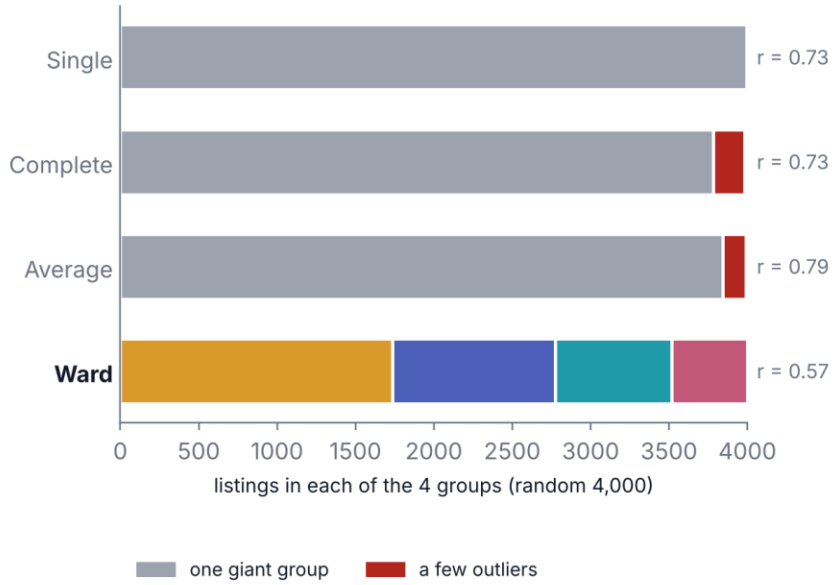
1 Hierarchical clustering → 2 K-means → 3 Comparison · bootstrap

Ward's tree on 2,000 random listings



Linkage methods

group sizes at k = 4 · cophenetic r on the right



Clear cut at k=4

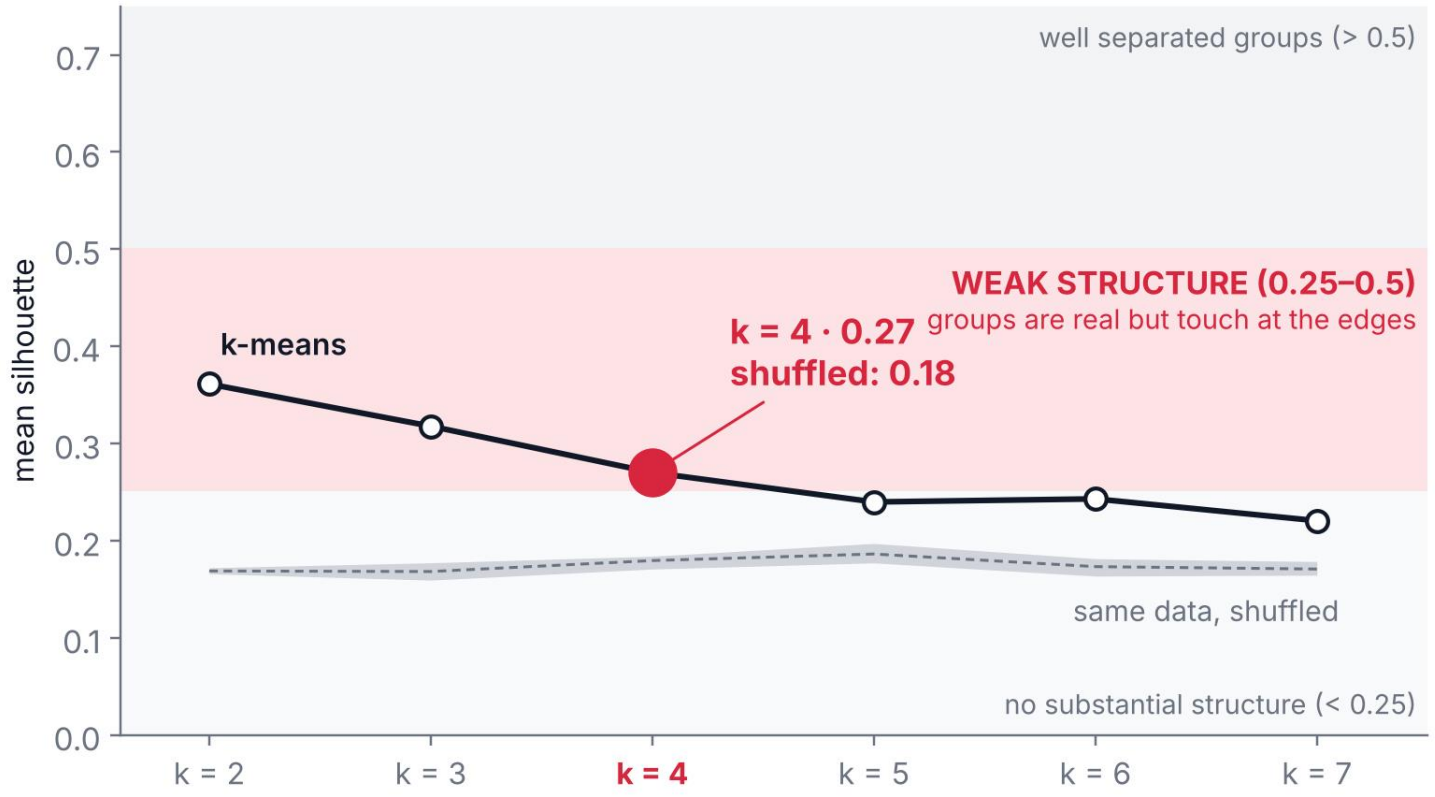
the merge height jumps above four branches, and each branch is one of the final types.

Only Ward is balanced

the others put 95%+ of listings in one group.

Q1

1 Hierarchical clustering → **2 K-means** → 3 Comparison · bootstrap



0.27

silhouette at k = 4

Real groups

gap between k-means and shuffled at every k → groups are in the data, not made up by the algorithm

Clear cores, soft borders

0.27 is inside the weak-structure band: each group has a clear core, but at the edges some listings fit two types almost equally well.

Q1

1 Hierarchical clustering → 2 K-means → **3 Comparison · bootstrap**

ARI (Adjusted Rand Index): compares two groupings pair by pair, corrected for chance.

k = 4

Both agree on k = 4

- Ward's tree cuts at 4
- k-means is very stable at 4

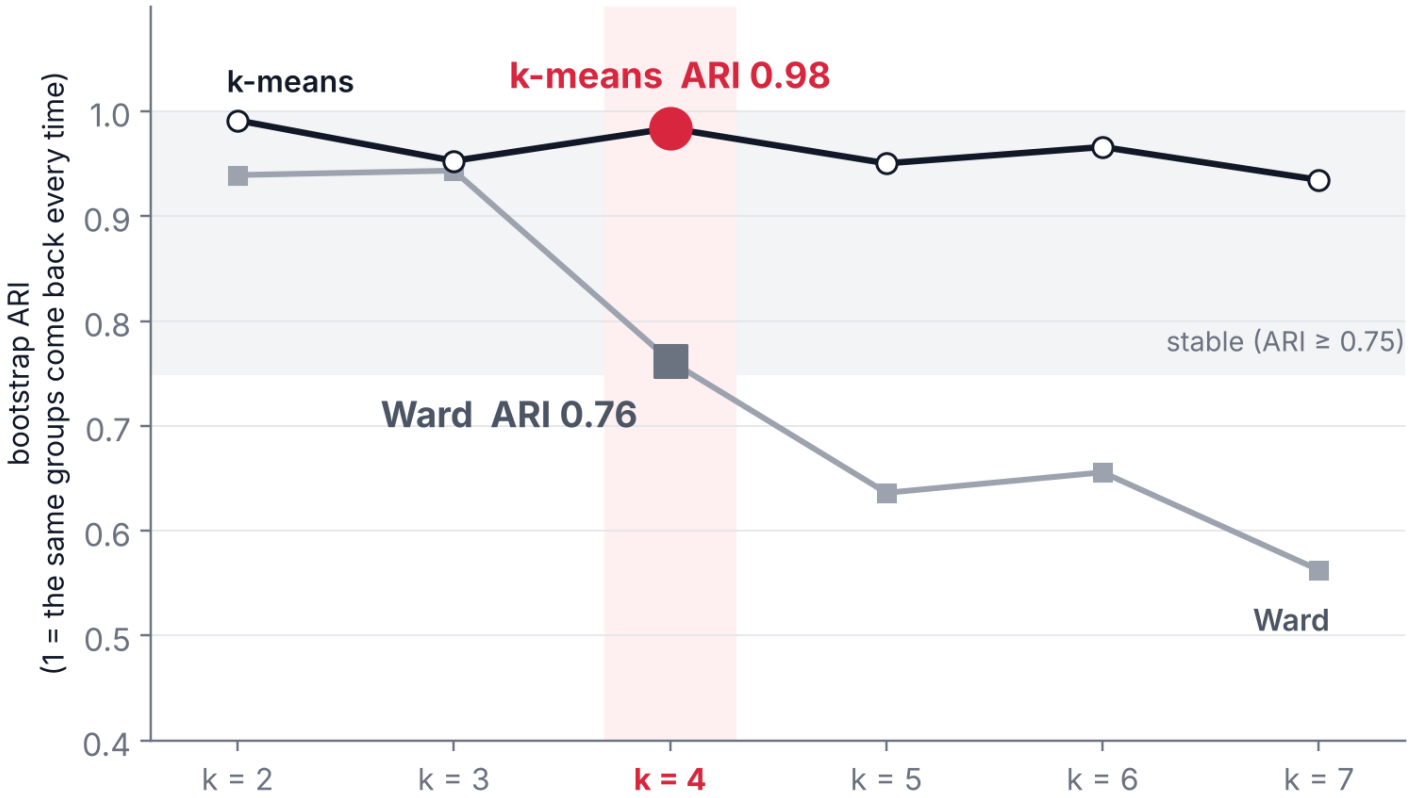
They find the same groups

Ward vs k-means at k = 4: ARI 0.87

K-means builds the final clusters

ARI: 0.76 for Ward vs 0.98 for k-means

→ k-means used to create clusters



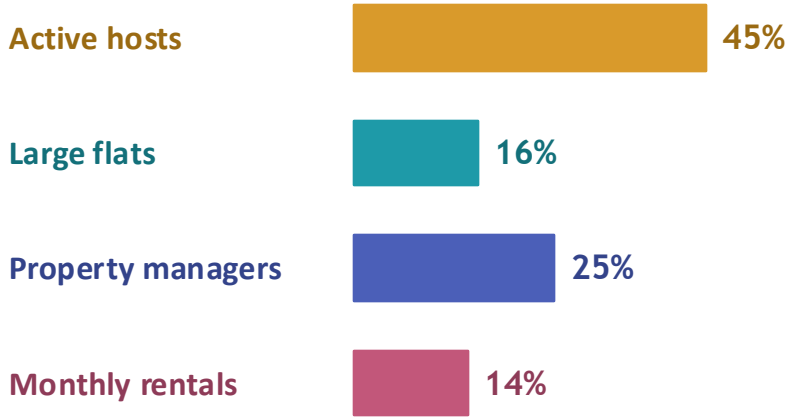
THE RESULT

PCA map

Q1



SHARE OF LISTINGS



Two types overlap

property managers differ from active hosts on PC3 (host scale), which this view does not show



What is each type like?



Description of the types

Q2

Active hosts

45% · 9,580 listings

Ordinary people renting one home, and renting it often.

1	listing per host
12	reviews a year
147 €	per night
45%	Superhosts

Large flats

16% · 3,374 listings

Big homes for families and groups.

6	guests
2	bedrooms
266 €	per night
39%	within 2 km of Duomo

Property managers

25% · 5,463 listings

Companies running dozens of flats, each booked little.

47	listings per host
2	reviews a year
190 €	per night
18%	Superhosts

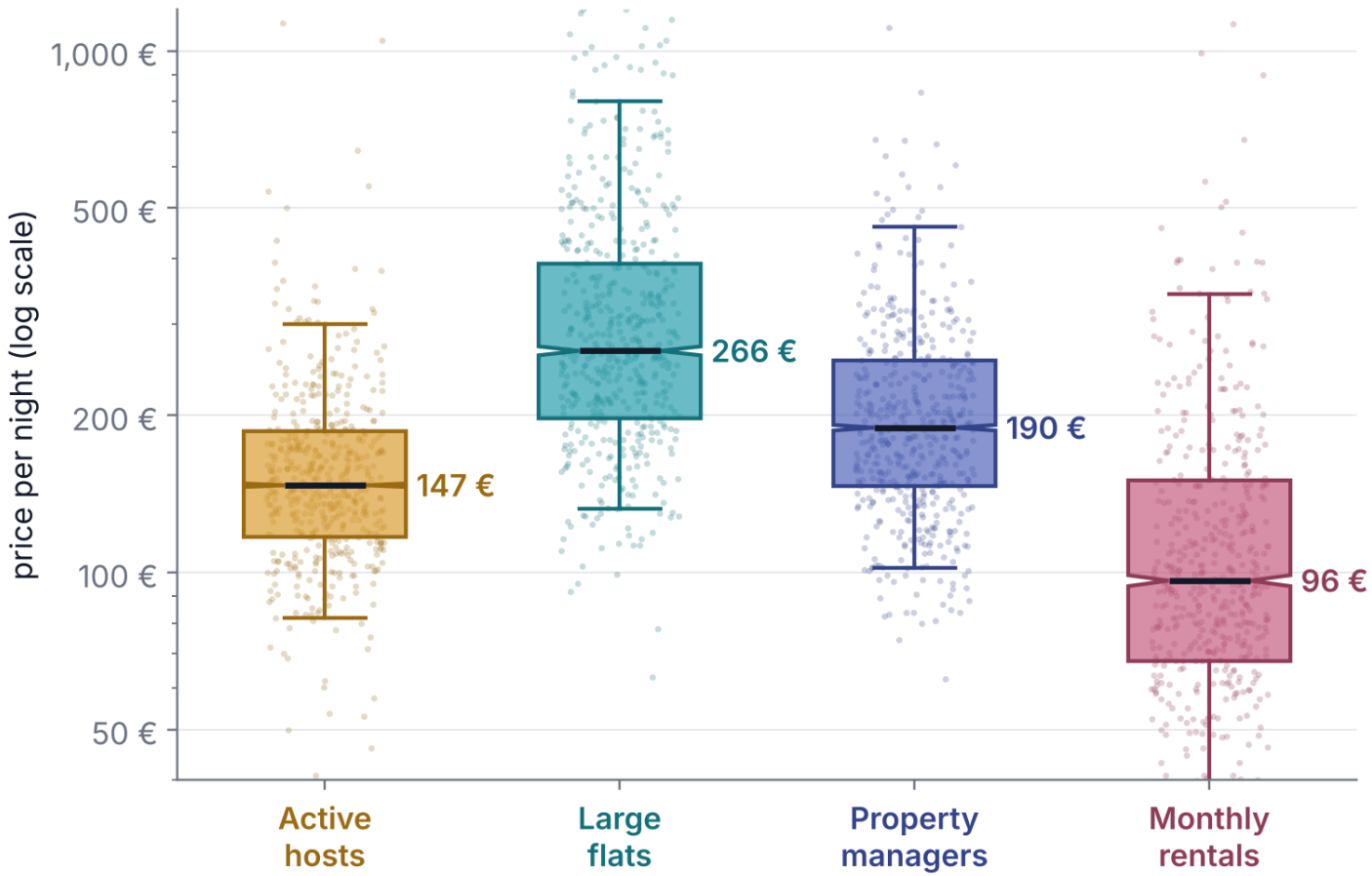
Monthly rentals

14% · 3,093 listings

Rooms and small flats for students and workers, by the month.

31	nights minimum
0	reviews a year
96 €	per night
29%	show the CIN code

Q2



266 € vs 96 €

median price per night, large flats vs monthly rentals

Notches do not overlap
the four medians are clearly different

Descriptive only
price helped build the types, so this is not a test of them



Do the types differ in hosting quality and licence compliance?



TESTS

Testing the types

We tested all 7 held-out variables (6 tests: latitude and longitude become one distance).

1

H_0

the 4 types do not differ on this variable

→

2

Test

χ^2 for categories, Kruskal–Wallis for skewed numbers

→

3

Result

p-value corrected for all tests run (Benjamini–Hochberg)

→

4

Pairs

which of the 6 pairs of types differ (Holm)

→

5

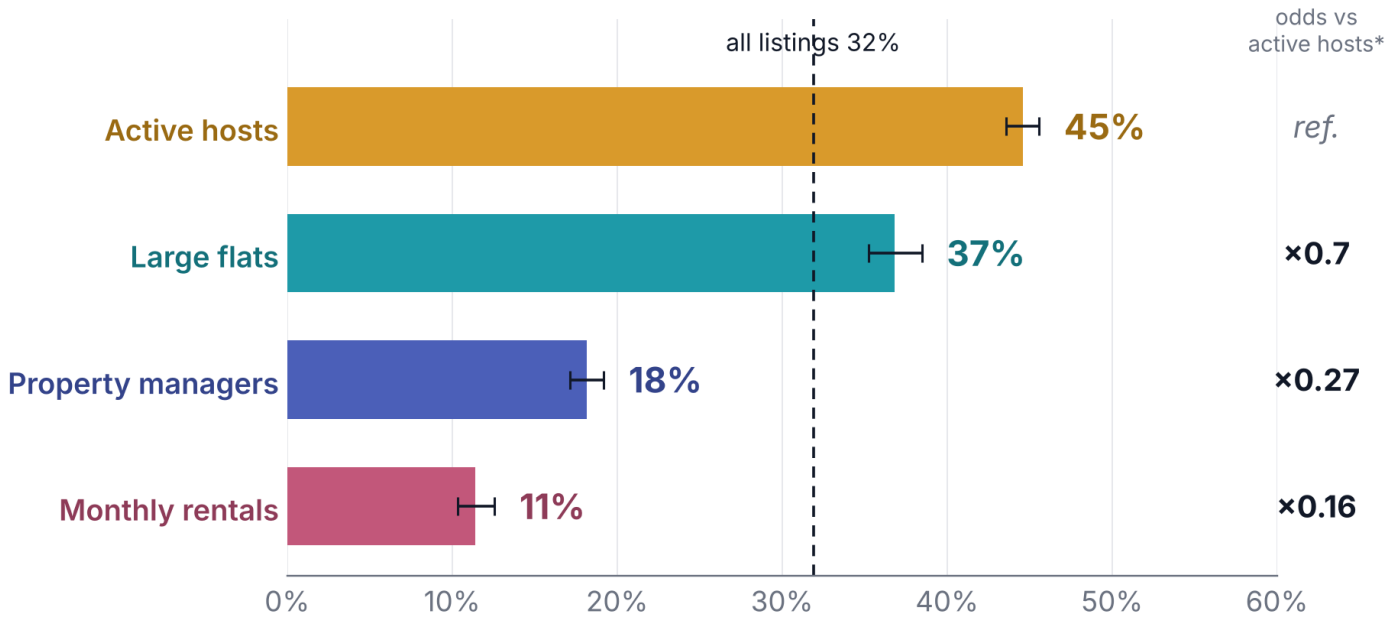
Size

Cramér V or ε^2: how big is the difference?

TEST 1

Superhost badge

Q3



* same distance from the Duomo and room type (logistic regression, errors clustered by host)

Confirms the types → the badge goes to people, not companies

H_0
same share of Superhosts in all 4 types

Test
χ^2 test, then logistic regression with distance and room type

Result
$p < 10^{-300}$ → reject H_0

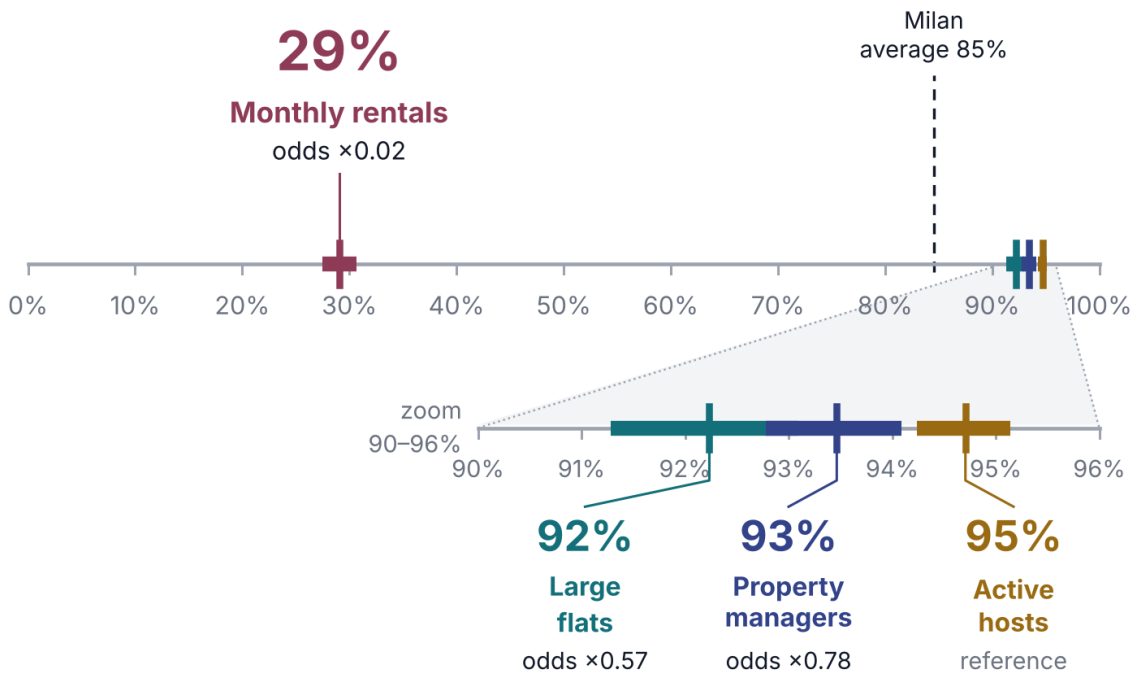
Pairs
all 6 differ (Holm)

Size
Cramér V = 0.29: small, close to medium

TEST 2

CIN licence code

Q3



tick = share of listings of that type showing the CIN code · thick segment = 95% CI
odds = vs active hosts at the same distance from the Duomo and room type

Confirms the types → monthly rentals fall outside the licence rule

H_0
same share showing the code in all 4 types

Test
χ^2 test, then logistic regression with distance and room type

Result
$p < 10^{-300}$ → reject H_0

Pairs
all 6 differ (Holm)

Size
Cramér V = 0.63: large, the biggest of all



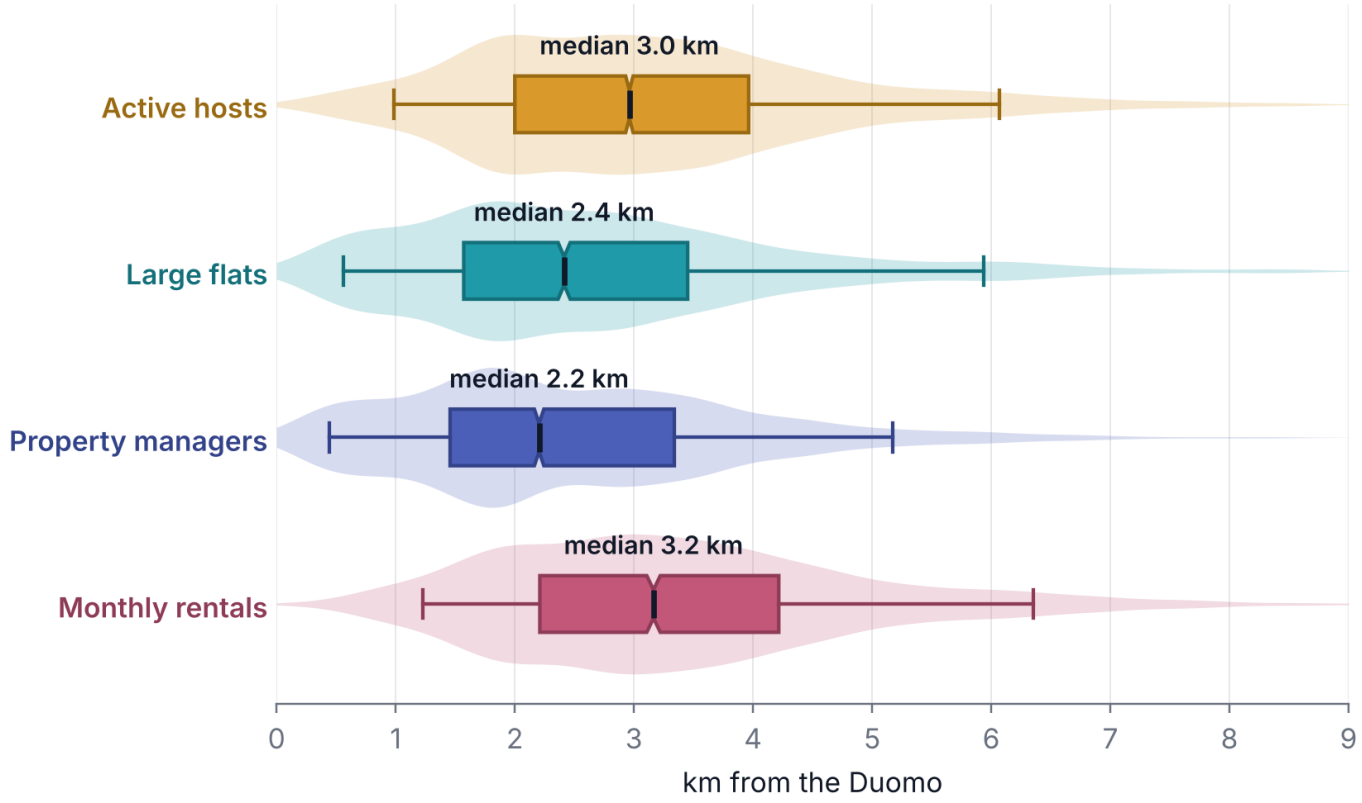
Are the types spread differently across Milan?



Distance from the Duomo

managers closest, rentals farthest

Q4



H_0
same distance for all 4 types

Test
Kruskal–Wallis (distances are skewed)

Result
$p < 10^{-243}$ → reject H_0

Pairs
all 6 differ (Holm); notches never overlap

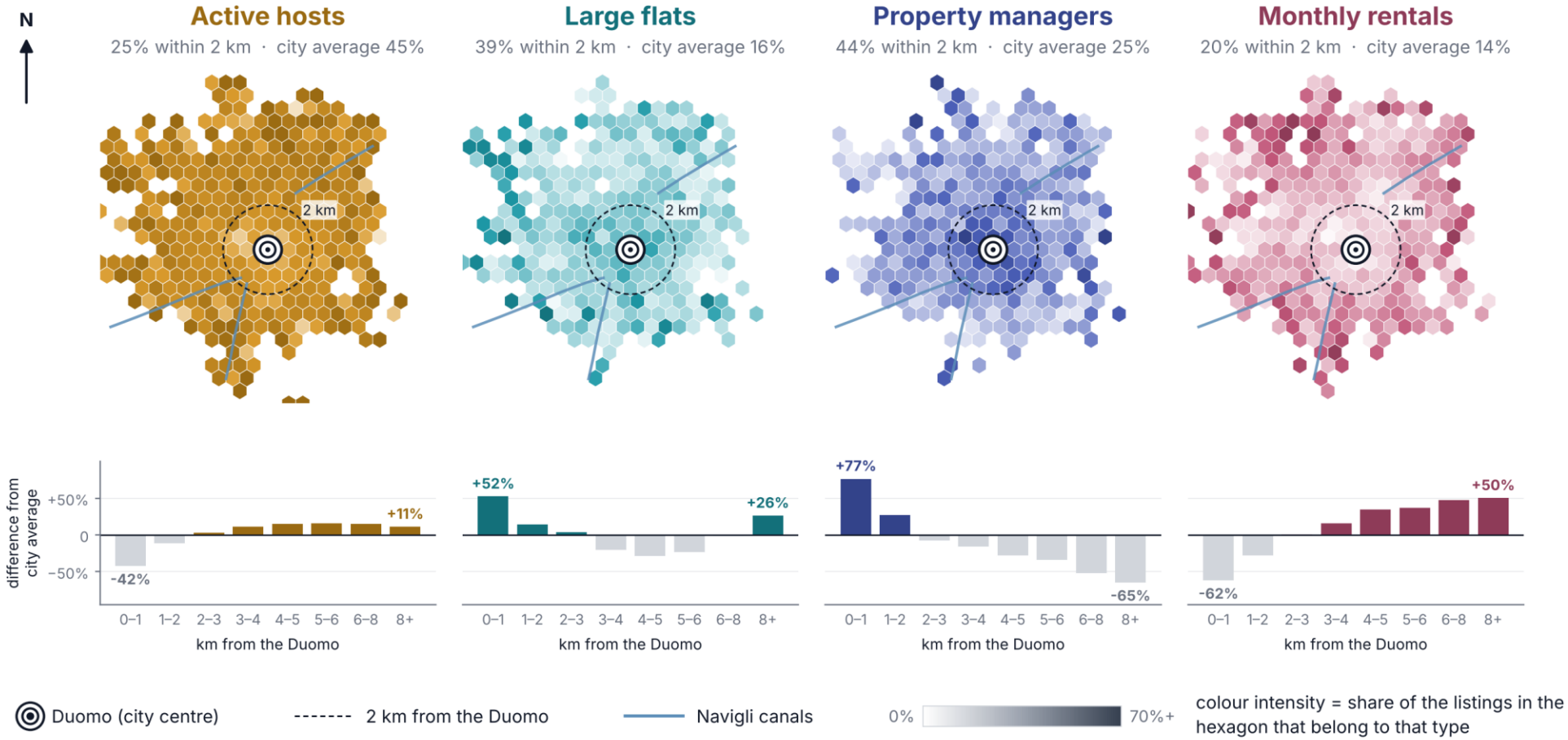
Size
$\varepsilon^2 = 0.05$: small → reasonable: groups shift by area, they don't split the city.

TEST 4

Neighbourhood

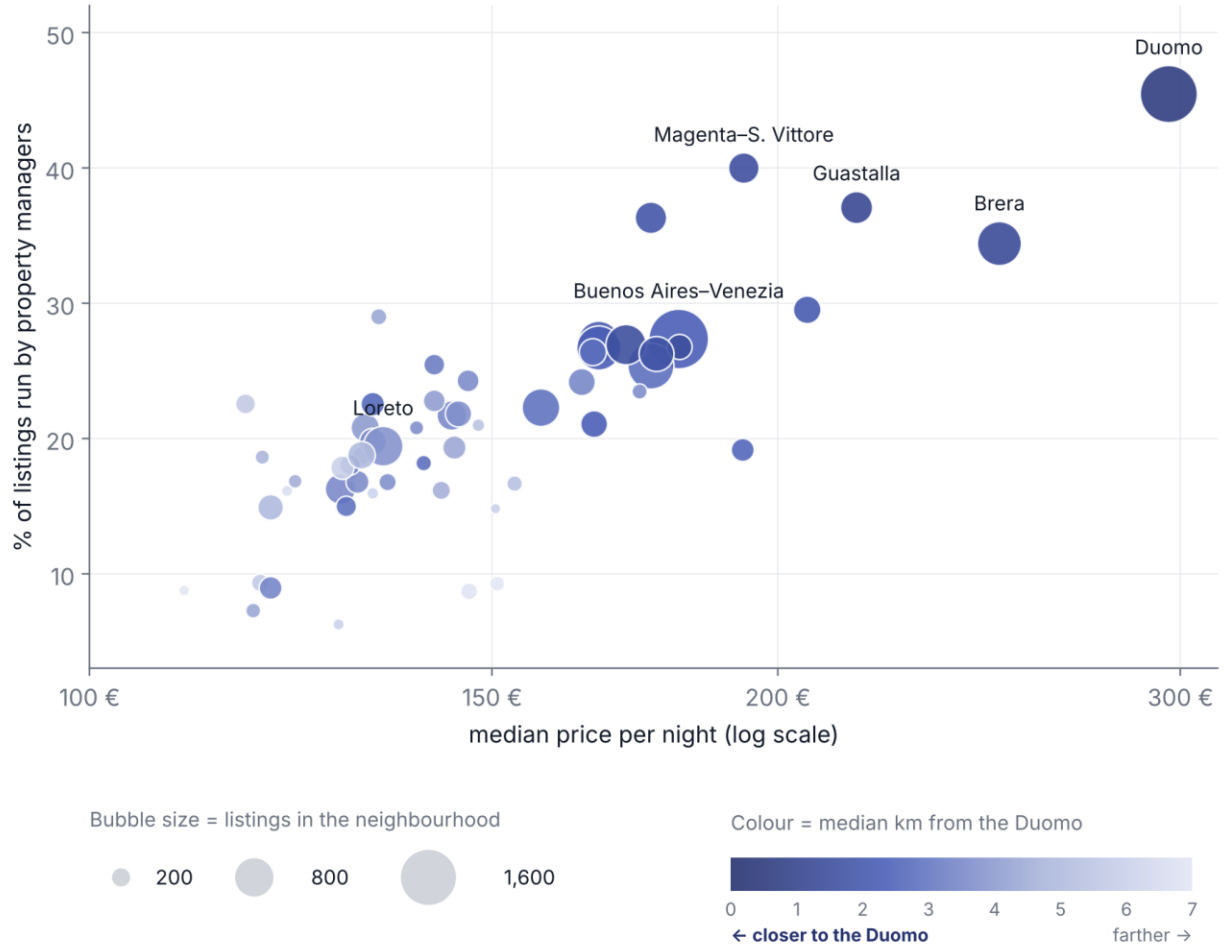
managers fill the centre, rentals the edge

Q4



Q4

The pricier it is, the more professional the hosts



$\rho = 0.73$

→ **the more expensive a neighbourhood, the larger the share of managers**

Duomo stands alone
297 € a night, 45% managers

The edge is cheap and private
below 150 € and under 20% managers

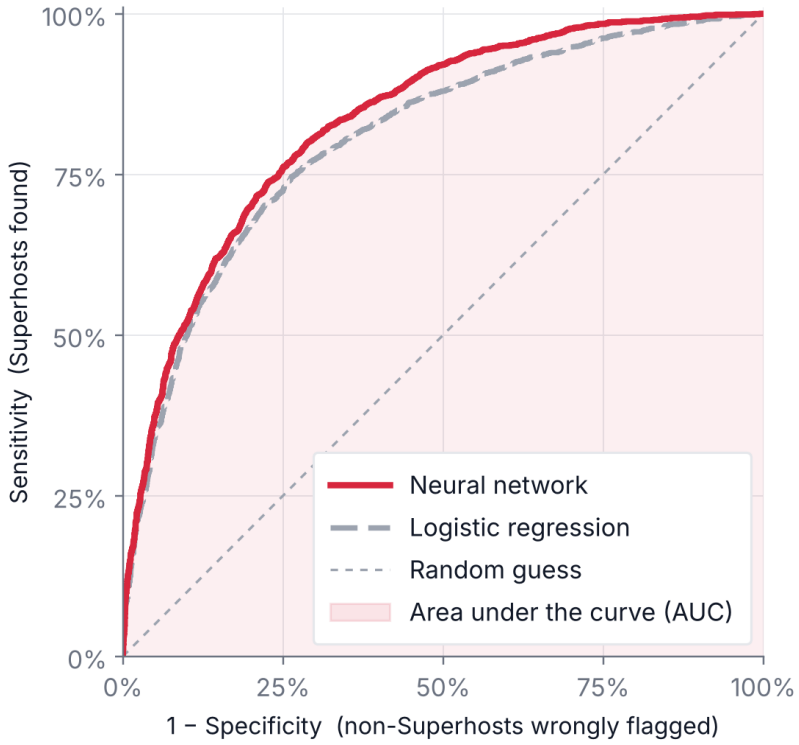


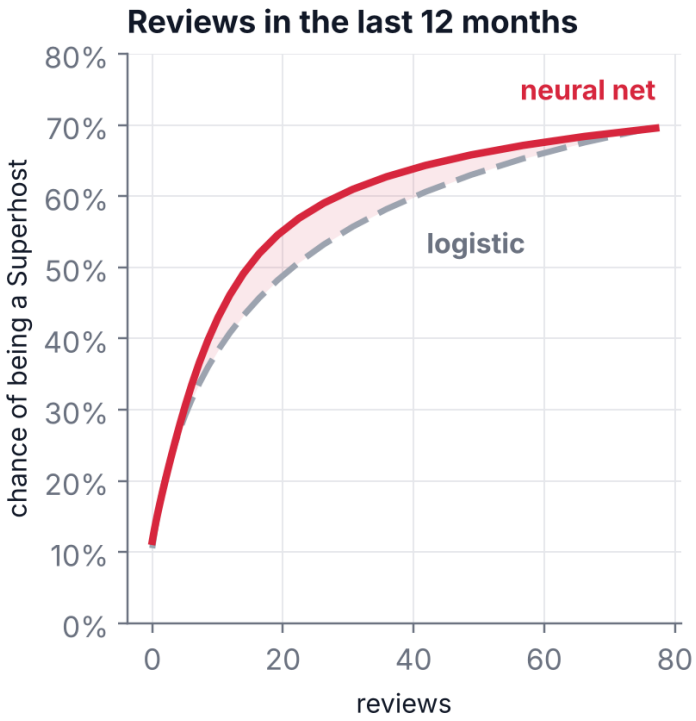
Can the listing alone predict a Superhost?

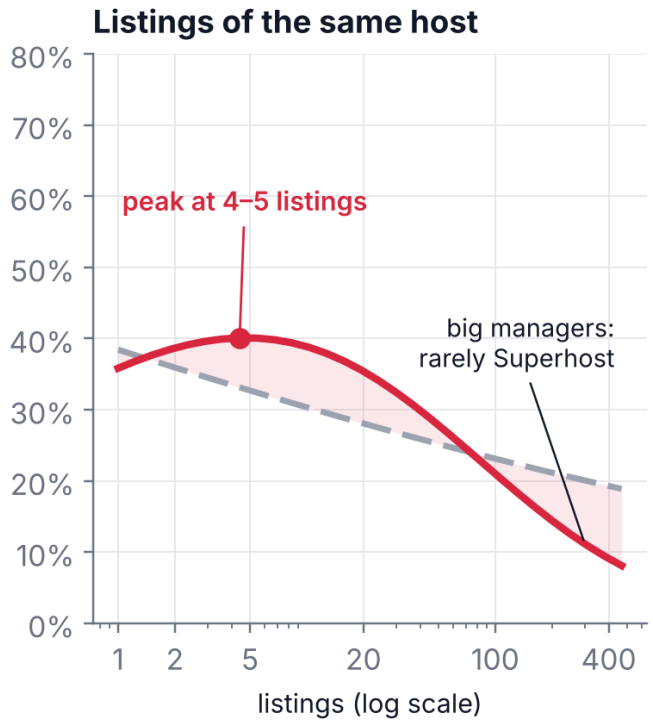


Q5

Predicting Superhosts: neural network and logit







AUC 0.83 vs 0.81 logit

AUC is the probability that the model ranks a random Superhost above a random non-Superhost.
Gain +0.025, 95% CI 0.006–0.048 (bootstrap by host)

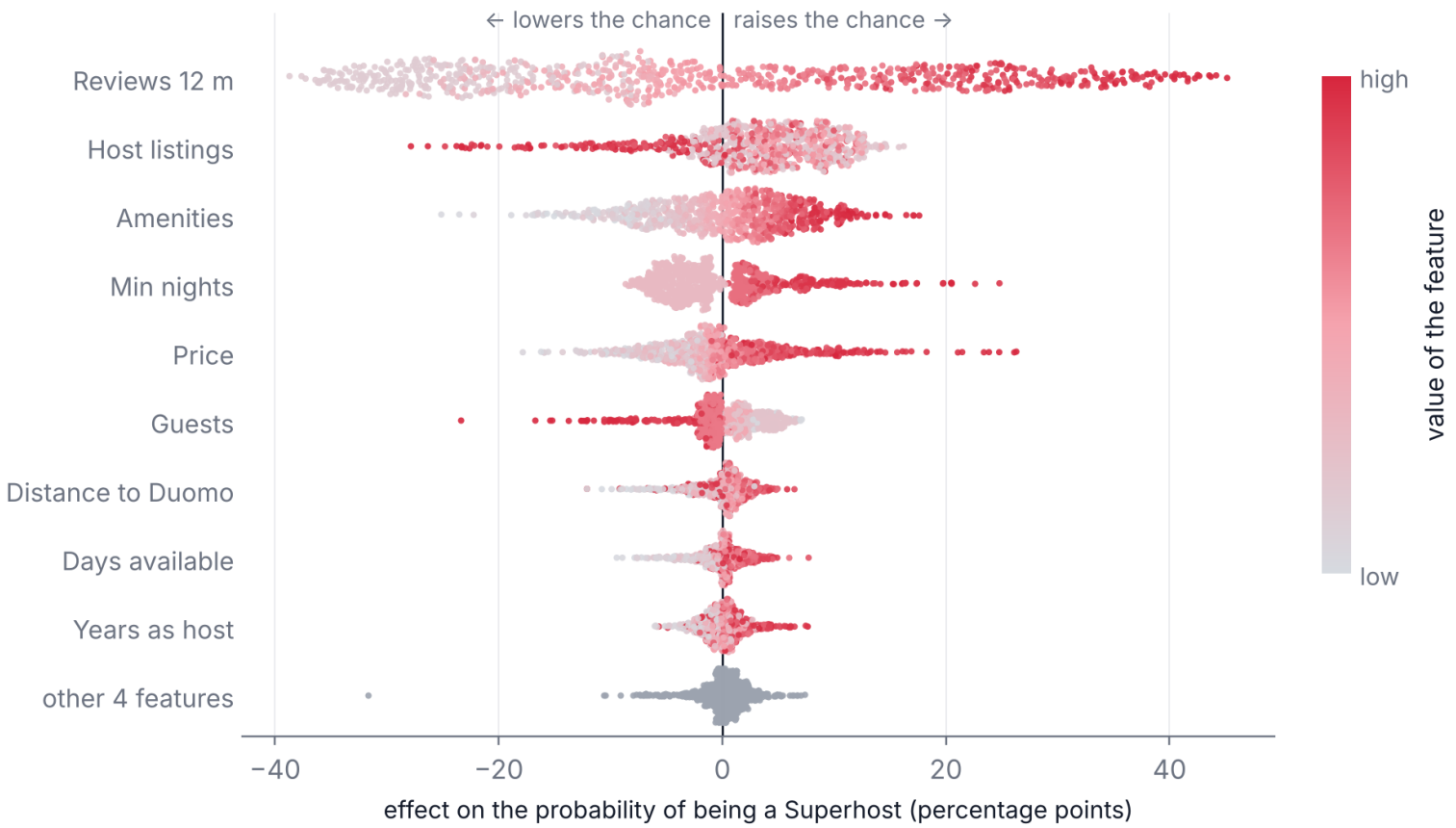
Why the net wins

Reviews: the chance of being a superhost climbs fast up to 20 reviews a year, then levels off.

Listings of the host: it peaks at 4–5 and falls for big managers, while the logit can only draw a straight line.

What drives the Superhost prediction

Q5



Reviews

matter most: many recent reviews add up to +40 points

Host listings

not a Superhost requirement → most interesting: the more listings a host has, the lower the probability (down to –28 points)

Up or down

amenities, price and minimum nights are positively associated with the Superhost chance; guests and bedrooms work the other way.

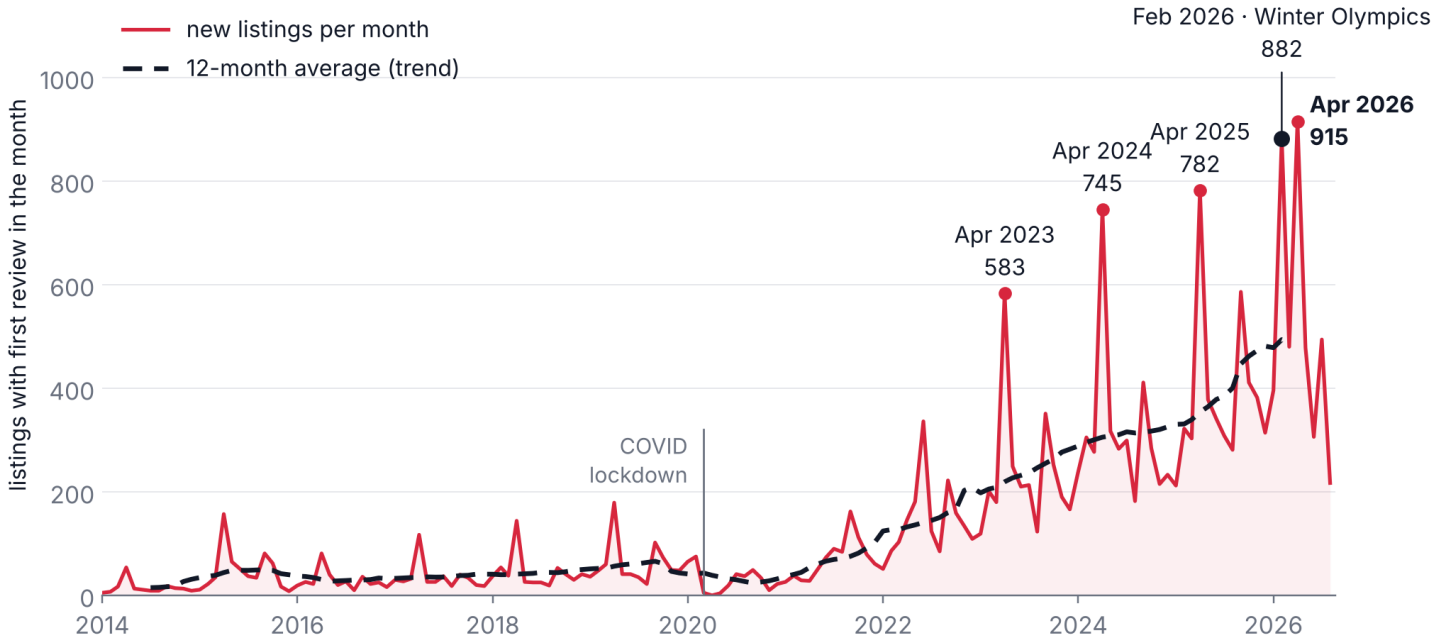


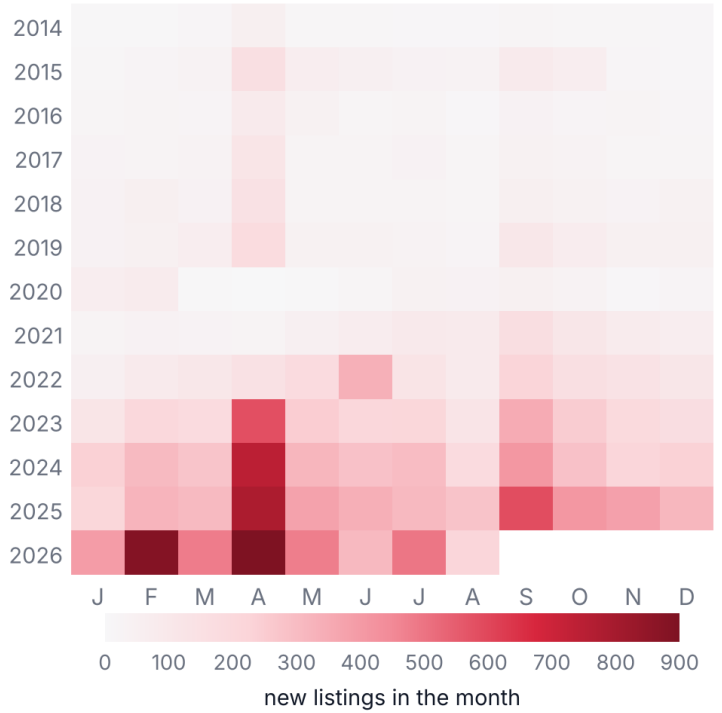
Is the market growing and who drives it?

OVER TIME

A young market that keeps growing

Q6





April

Milan Design Week
Record in April 2026: 915.

81%

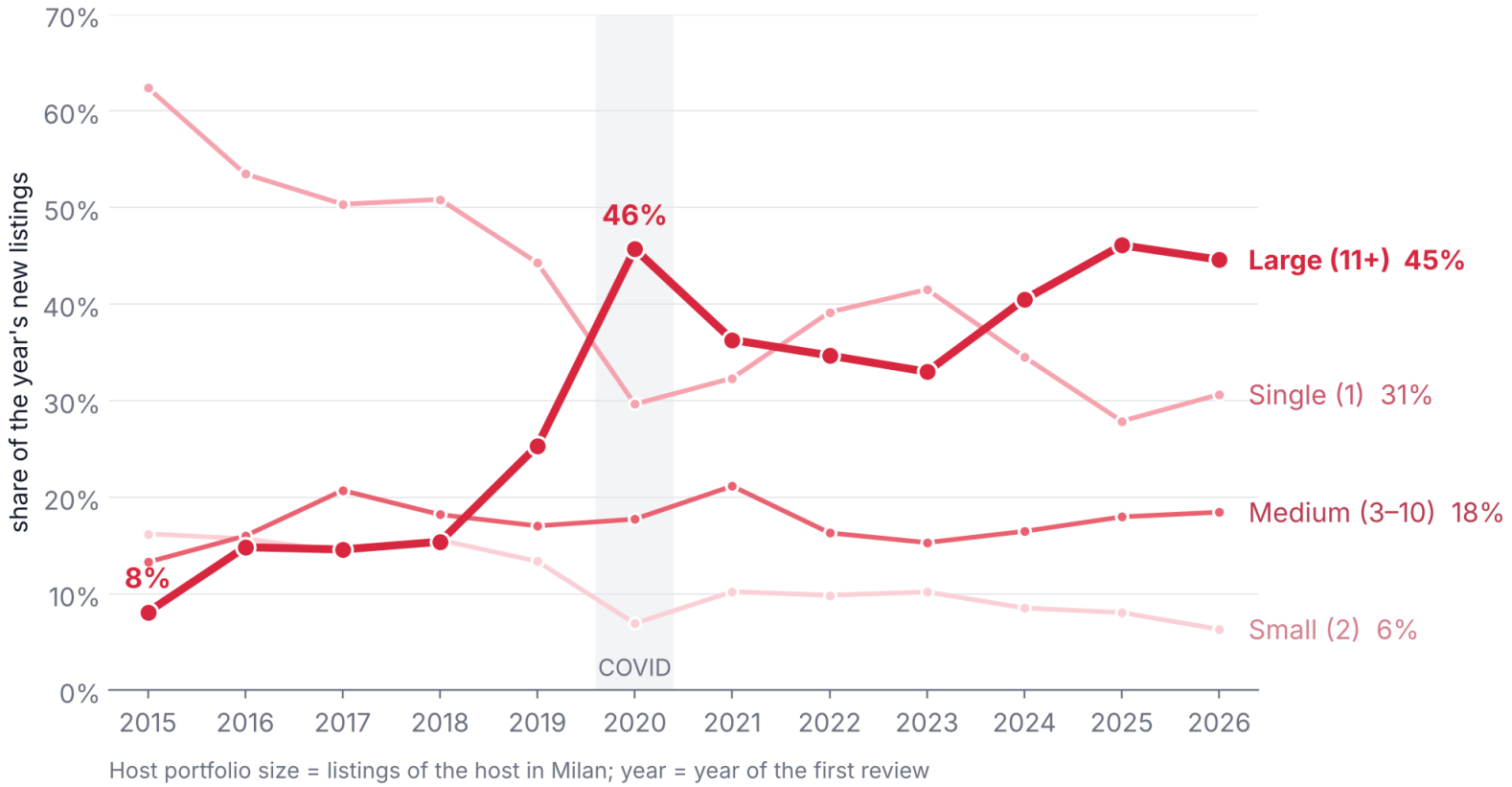
of today's listings entered the market in 2022 or later.

×2.7

new listings in February 2026 (Winter Olympics) vs February 2025: 882 vs 322.

Who publishes new listings?

Q6



Host portfolio size = listings of the host in Milan; year = year of the first review

8% → 45%

share of new listings from large operators, 2015 → 2026

2020: the professionals kept going
In the COVID year single hosts stopped listing, professionals did not (25% → 46%)

Almost 1 new listing in 2
now comes from a host with 11+ listings: the same professionals who crowd the centre.

CONCLUSION

Answers and limitations

Q1 Four types, real and stable, with soft borders

Q2 Active hosts, large flats, managers, monthly rentals

Q3 Managers are rarely superhosts, monthly rentals not always licensed

Q4 Managers in the centre, rentals at the edge

Q5 The neural net beats the logit

Q6 Peaks with big events, driven by professionals

LIMITS

One day only

a snapshot of 21 September 2026

Reviews, not bookings

not every guest leaves a review

Asking prices

what host ask, not what guests paid

Hosts not independent

host-based split

Who really rents Milan on Airbnb?

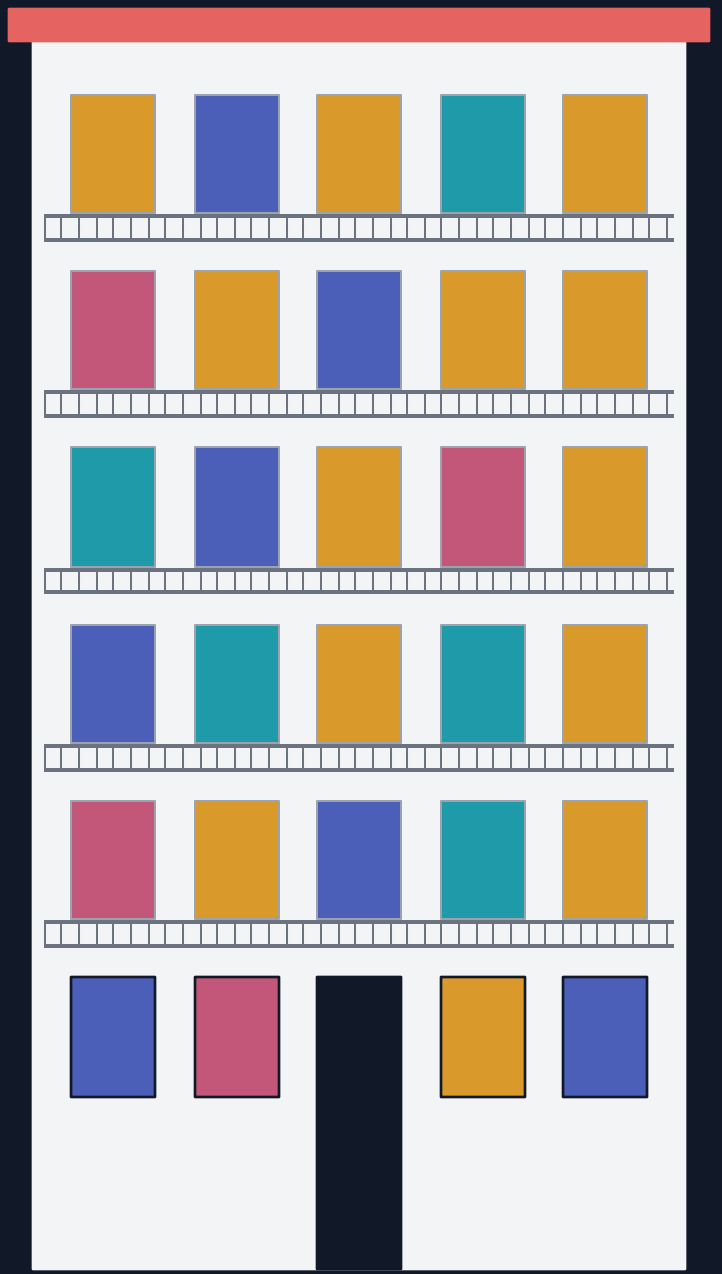
Four kinds of listing, and the professionals are taking over.

1 in 4

listings is run by a property manager

45%

of 2026's new listings came from large operators



Thank you